



COLLEGE
OF
MEDICINE

Family planning and Contraception

Dr. ROZHAN YASSIN KHALIL

Assistant professor & consultant obstetrics & gynecology
FICOG,CABOG,HDOG,FICS,MBChB

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ROZHAN YASSIN KHALIL



Objective:

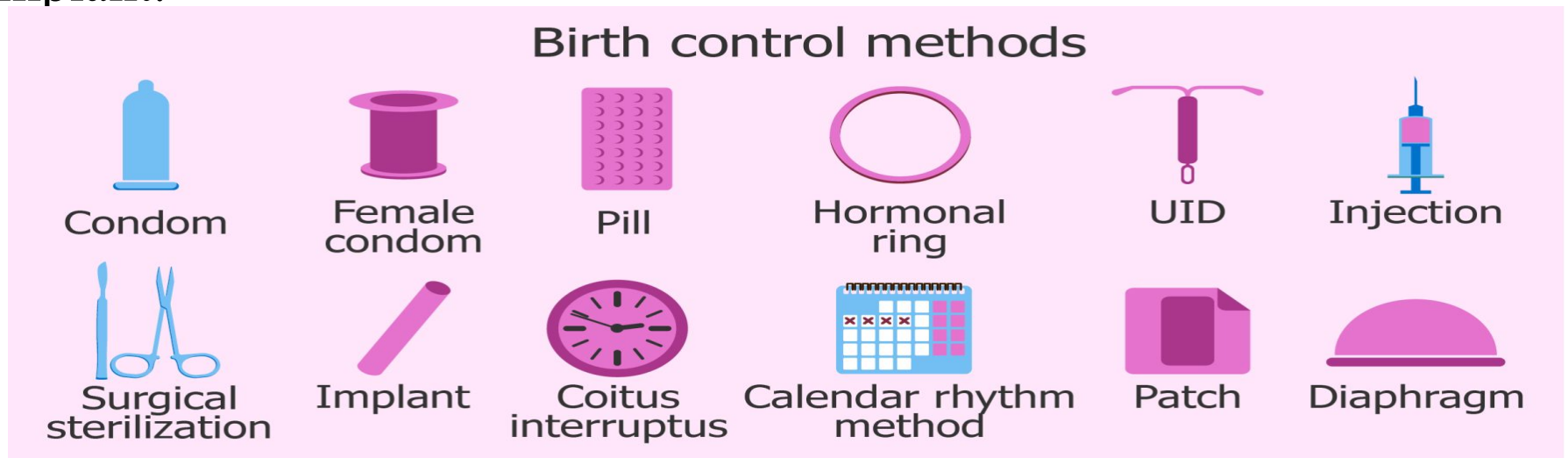
- Types of contraceptions.
- How to choose types of contraception.
- Mechanism of action of contraception.
- Failure rate of contraception.
- Side effects of contraception.



- Abortion is a major consequence of unintended pregnancy.
- In order to prevent pregnancy, contraceptive methods could be developed to target one or a number of key reproductive processes or sites in the male or female reproductive tract.
- None of the existing methods of contraception are **100%** effective at preventing pregnancy.
- The effectiveness of a method depends on both mechanism of action and correct and consistent use.
- Compliance depends on acceptability of the method to the user and tolerability with any untoward effects that they experience related to use of the method.

A woman's choice of contraception is just as likely to be based upon information from media, friends and family, as from a health care professional.

The most effective methods of contraception are the long acting reversible methods of contraception (LARC) – or so called '**fit and forget methods**' such as the copper intrauterine device (Cu-IUD), levonorgestrel intrauterine system (LNG-IUS) and progestogen-only implant.





The characteristics of ideal contraceptive method would be:

- Highly effective
- No side effects or risks .
- Cheap.
- Independent of intercourse and requires no regular action on the part of the user.
- Have non- contraceptive benefits.
- Acceptable to all cultures and religions.
- Easily distributed and administrated by non-healthcare personnel.
- Quick return to fertility after cessation of contraception



Mechanism of action of contraceptions

- **Prevent ovulation:** this is the mechanism of action of the following methods: combined hormonal methods (pill, patch and vaginal ring), progestogen-only injectables, progestogen-only implant (Nexplanon®), oral emergency contraception, lactational amenorrhoea.

Prevent sperm reaching the oocyte: female sterilization and male sterilization (vasectomy).

- **Prevent an embryo implanting in the uterus:** this is a mechanism of action of the Cu-IUD and LNG-IUS.



- *Allow sperm into the vagina but poison them*: mechanism of action of spermicides.
- *Allow sperm into the vagina but block further passage*: mechanism of action of diaphragm and cap. Also one of the mechanisms of action of progestogens.
- *Prevent sperm entering the vagina*: male and female condoms;
avoid sex during the fertile time of the cycle and fertility awareness-based methods (FAB).

- Real-life effectiveness depends on compliance and continuation with the method. Compliance is influenced by the route of administration; some methods are easier to use than others.
- Continuation with a method depends on the acceptability to the user.
- **Failure rates** are expressed as the number of pregnancies if 100 women were to use a method for one year.
- Failure rates during perfect use show how effective methods can be, where perfect use is defined as following the directions for use.
- Failure rates during typical use show how effective the different methods are during actual use (including inconsistent or incorrect use).



Percentage of women experiencing an unintended pregnancy within the first year of use with typical use and perfect use

Method	Typical use %	Perfect use %
No method	85	85
Fertility awareness-based methods	24	0.4–0.5
Male condom	18	2
Female diaphragm	12	6
Progestogen-only pill	9	0.3
Combined hormonal contraception*	9	0.3
Progestogen-only injectable	6	0.2
Cu-IUD	0.8	0.6
LNG-IUS	0.2	0.2
Progestogen-only implant	0.05	0.05
Female sterilization	0.5	0.5
Vasectomy	0.15	0.1



- **LARC have been defined in the UK as a method that requires administration less than once per month (i.e injectable, implant, intrauterine device [IUD]), although in many other countries, the injectable is not considered as LARC, but as a 'medium acting' method.**
- **LARC methods are the most effective as once inserted they do not require any action by the user until they need to be renewed (3 years for the implant Nexplanon® and 5–10 years or longer for IUDs).**

- In contrast, the shorter-acting methods do require compliance by the user (e.g. to take a daily pill, change a weekly patch or use a condom for every act of sex).
- LARC methods have ‘typical’ failure rates within the first 12 months of use that are similar to those of ‘perfect’ failure rates. In contrast, for short-acting methods typical failure rates are much higher than perfect failure rates.



Safety:

- Most women who use contraception are fit and healthy. However, some health conditions may be associated with real or theoretical risks if a particular contraceptive method affects the health condition.
- WHO developed a system addressing medical eligibility criteria for contraceptive use.

**MEC
category**

Definition of category

1

A condition for which there is no restriction for the use of the contraceptive method

2

A condition where the advantages of the method generally outweigh the theoretical or proven risks

3

A condition where the theoretical or proven risks generally outweigh the advantages of using the method
The provision of a method requires expert clinical judgement and/or referral to a specialist contraceptive provider, since use of the method is not usually recommended unless other more appropriate methods are not available or not acceptable

4

A condition that represents an unacceptable health risk if the contraceptive method is used



Examples of WHO medical eligibility criteria category 4 conditions and use of combined hormonal COI

Age >35 and smoking

Blood pressure >160/100 mmHg

Hypertension with vascular disease

Deep vein thrombosis, current or past

Myocardial infarction, current or past

Cerebrovascular accident, current or past

Multiple serious risk factors for cardiovascular
disease

Known thrombogenic mutations

Current breast cancer



Interaction with other medicines

- There are a number of medicines (some anticonvulsants, antifungals, antiretrovirals and antibiotics) that induce liver enzymes cytochrome P450, and will reduce the efficacy of hormonal contraception such as CHC pills, patch or ring, progestogen-only implant and progestogen-only pill (POP).
- If a woman using enzyme-inducing medication wishes to use one of these hormonal methods, then the consistent use of condoms is also advised.

- Alternatively, she could consider use of the progestogen-only injectable, Cu-IUD or LNG-IUS, since efficacy of these methods is not affected by drugs that are enzyme inducers.
- Effectiveness of the combined oral contraceptive pill (COCP) (and all other methods) is not affected by administration of most broad-spectrum antibiotics.



Type of drug	Liver enzyme induction
Anticonvulsant	Carbamazepine Eslicarbazepine Oxcarbazepine Phenobarbital Phenytoin Primidone Topiramate
Antibiotic	Rifampicin Rifabutin
Antifungal	Griseofulvin
	Protease inhibitors Amprenavir Atazanavir Nelfinavir Lopinavir
Antiretroviral	Saquinavir Ritonavir Non-nucleoside reverse transcriptase inhibitors Efavirenz Nevirapine

Side-effects

- Common side-effects that women report with all hormonal methods are unexpected bleeding, weight gain, headaches, mood swings and loss of libido.
- Unexpected bleeding is common (15%) when women start a COCP and may settle with time. If not, there is some evidence that changing to a different contraceptive pill with a different dose of hormones may help.
- If bleeding problems persist for more than 3 months, investigations should be done to exclude other causes **(e.g. cervical conditions such as polyps, chlamydial infection or intrauterine cavity lesions such as submucous fibroids or polyps).**



Non-contraceptive health benefits of hormonal contraception

Method	Benefit against
LNG-IUS (52 mg)	Heavy menstrual bleeding Endometriosis Adenomyosis Dysmenorrhoea Endometrial protection Simple hyperplasia
Combined hormonal contraception	Heavy menstrual bleeding Irregular menses Hirsutism Acne Premenstrual syndrome Reduces risk of ovarian cancer Reduces risk of endometrial cancer
Progestogen-only injectable (depot medroxyprogesterone acetate)	Heavy menstrual bleeding Endometriosis Dysmenorrhoea



Acceptability

This determines whether women choose to start a method and whether they continue to use that method.

Determinants of contraceptive method acceptability

- Personal characteristics (e.g. age).
- Fertility intentions.
- Perceptions of effectiveness.
- Perceptions of safety.
- Fear of side-effects.
- Familiarity.
- Experience of others.
- Ease of use and of access.
- Need to see a health professional.
- Intrusiveness.
- Non-contraceptive benefits.



What a woman needs to know before starting a method of contraception

- How to use the method (pill, patch or ring) and what to do when misused (e.g. missed pill)
- Typical failure rates
- Common side-effects
- Health benefits
- Fertility return on stopping
- When she requires review



- CHC methods contain two hormones: an oestrogen and a progestogen.
- They are available as oral pills, a transdermal patch and as a vaginal ring.
- They are similar in terms of effectiveness, safety and side-effects.
- These methods all work by inhibition of ovulation via negative feedback of oestrogen and progestogen on the pituitary, with suppression of FSH and LH.

Pills

Most of the commonly used COCPs are 'low dose' and contain ethinyl oestradiol in a dose of 15–35 µg. Some newer pills contain oestradiol valerate or oestradiol hemihydrate, which is more similar in structure to the 'naturally occurring' oestradiol,



The oral contraceptive pill.



- Most ‘traditional’ preparations contain 21 pills followed by a 7-day pill-free interval (or 7 placebo tablets in place of a 7-day pill-free interval). Some preparations contain 24 days of pills with a shorter pill-free interval.
- Preparations are commonly monophasic (i.e. same dose of hormones throughout), but some are phasic (dose varies).
- **There is no advantage of phasic preparations over monophasic preparations.** Although traditional 21 days on and 7 days off usually results in a withdrawal bleed during the pill-free interval.

- Women with dysmenorrhoea or headaches during the pill-free interval, are often advised to do so to avoid recurrence of symptoms during the hormone-free interval by tricycling (taking three packets without any break).
- The progestogens that are used in currently available pills are often referred to as second-generation (levonorgestrel, norethisterone), third-generation (gestodene desogestrel) and fourth-generation progestogens (drospirenone and dienogest).



- Newer (third-and fourth-generation) progestogens were developed to have advantages due to less androgenic activity, but seem to be associated with a higher risk of venous thrombosis than pills containing second-generation progestogens.
- In view of this, COCPs containing second-generation progestogens are generally recommended as first choice.

Patch and ring

- The combined hormonal transdermal patch releases 33.9 μg ethinyloestradiol/day and norelgestromin 203 μg /day.
- It is applied to the skin of the lower abdomen, buttock or arm for 7 days, although it can be applied to any skin covered area, except the breast .
- The regimen usually involves application of patches for a total of 21 days followed by a 7-day hormone-free interval.
- Continued use (tricycling or tailored use) is also possible.
- Some women may experience problems with patch adherence or skin sensitivity to the patch.

- The combined hormonal ring is a flexible ring of 54 mm diameter that releases 15 μg ethinyloestradiol and 120 μg etonorgestrel daily, and as such is the lowest dose combined hormonal method.
- The ring is self inserted and worn in the vagina for 21 days, followed by a 7-day hormone-free interval, during which a withdrawal bleeding occurs.
- Women should not feel discomfort from the ring and it can be removed for a short time (less than 3 hours) and can be cleaned and replaced.

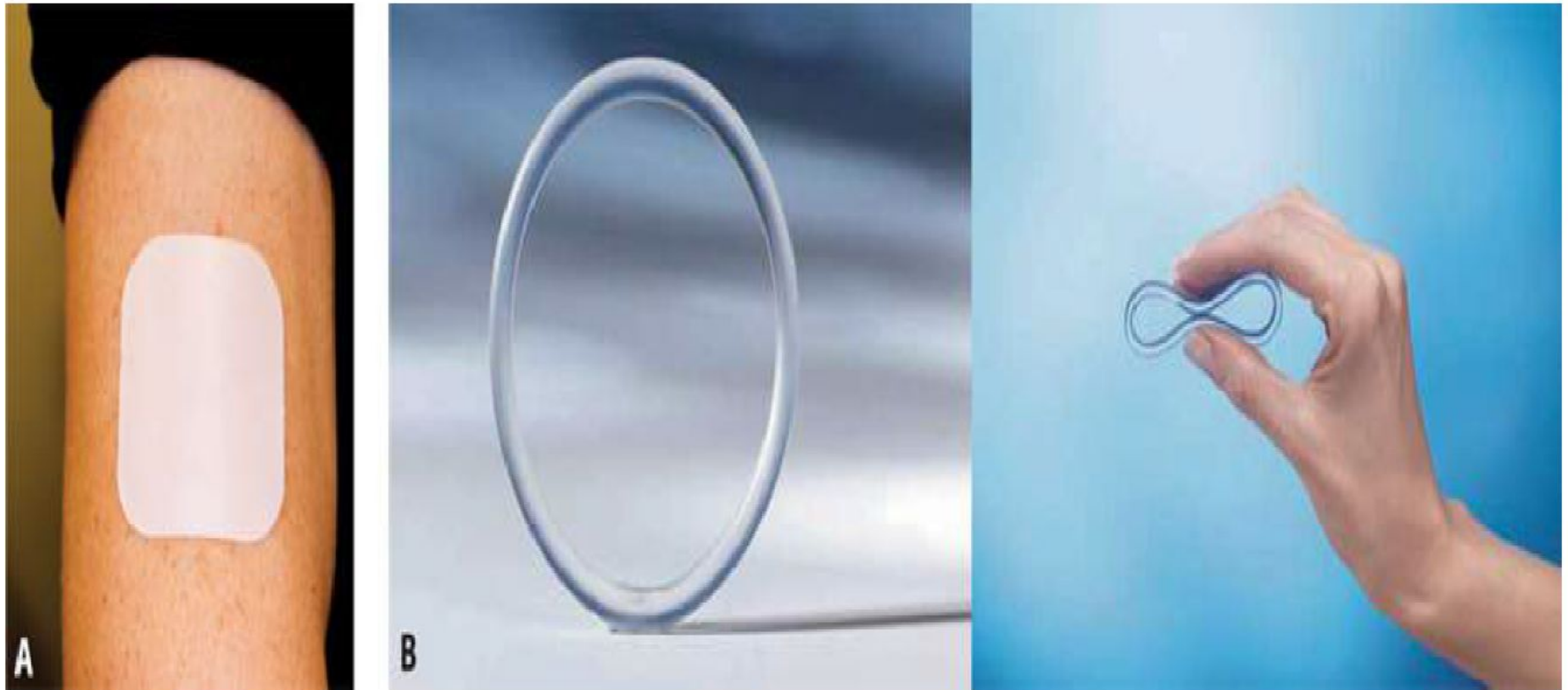


Figure 6.2 A: Combined hormonal patch. B: combined hormonal vaginal contraceptive ring.



Missed pills, patches and

rings

- Missed pills are common and if two or more are missed, then this places a woman at risk of ovulation.
- Additional contraceptive cover (condoms or abstinence) is required for most monophasic ethinyloestradiol-containing pills during the next 7 days of pill taking.
- Additional precautions are also required if a patch is not applied for 48 hours or a ring for more than 3 hours.
- If unprotected sexual intercourse has occurred during this time then there is a risk of pregnancy and so emergency contraception is recommended.

If one pill has been missed (more than 24 hours and up to 48 hours late)



Continuing contraceptive cover

The missed pill should be taken as soon as it is remembered

The remaining pills should be continued at the usual time



Minimizing the risk of pregnancy

Emergency contraception (EC) is not usually required but may need to be considered if pills have been missed earlier in the packet or in the last week of the previous packet

If two or more pills have been missed (more than 48 hours late)



Continuing contraceptive cover

The most recent missed pill should be taken as soon as possible

The remaining pills should be continued at the usual time

Condoms should be used or sex avoided until seven consecutive active pills have been taken. This advice may be overcautious in the second and third weeks, but the advice is a backup in the event that further pills are missed



Minimizing the risk of pregnancy

If pills are missed in the first week (pills 1–7)

EC should be considered if unprotected sex occurred in the pill-free interval or in the first week of pill taking

If pills are missed in the second week (pills 8–14)

No indication for EC if the pills in the preceding 7 days have been taken consistently and correctly (assuming the pills thereafter are taken correctly and additional contraceptive precautions are used)

If pills are missed in the third week (pills 15–21)

OMIT THE PILL-FREE INTERVAL by finishing the pills in the current pack (or discarding any placebo tablets) and starting a new pack the next day

How to manage missed pills



Cancer risks among users of COCPs

- A 12% reduction in the risk of any cancer.
- Reduced risk of colorectal cancer.
- Reduced risk of endometrial cancer.
- Reduced risk of ovarian cancer.
- Increased risk of breast cancer during use (decreases on stopping and similar risk to never used by 10 years after stopping).
- Increased risk of cervical cancer (but early changes detected by cervical cytology and human papillomavirus [HPV] vaccination).



Venous thromboembolism and arterial disease

- CHC (pill, patch and ring) increases the tendency to thrombosis in both the venous and arterial circulation.
- The adverse effect on venous thrombosis is related to the dose of oestrogen and appears less with combined pills containing second-generation progestogens compared to those containing third- or fourth-generation progestogens.
- However, whichever progestogen is used, the absolute risk of venous thromboembolism (VTE) is very small and much less than that associated with pregnancy

Risk of VTE in users and non-users of CHC

- 5 per 10,000 in non-pregnant non users.
 - 10 per 10,000 COCP users.
 - 29–400 per 10,000 in pregnant/postpartum.
-
- The risk of VTE is greatest during the first year of use, possibly due to the unmasking of inherited thrombophilias.
 - Women should be asked about a personal and family history of VTE if considering using this method, as these are contraindications to using a CHC method.
 - Women who are using CHC and making long-distance travel (>3 h of immobility) should take appropriate exercise on the journey and consider wearing graduated compression socks. Arterial disease is much less common but more serious.



- It is related to age, and the risk is strongly influenced by smoking.
- Women over 35 years old who smoke are not eligible to use a CHC method.
- The combined pill is also contraindicated in women who experience migraine with aura (homonymous hemianopia, unilateral paraesthesia, weakness, aphasia or unclassifiable speech disorder occurring before the headache), since this condition is associated with cerebral vasospasm and women may be at higher risk of stroke if they use a CHC.

Progestogen-only contraceptive methods:

- Are available as oral, injectable, implant and intrauterine system.
- The mechanism of action of the method and the bleeding pattern appear to depend on the dose of progestogen and also the route of administration.
- The injectable, implant and desogestrel-containing POP inhibit ovulation.
- Lower-dose POP formulations inhibit ovulation only inconsistently.

- All progestogen-only contraceptive methods, regardless of the route of administration, thicken cervical mucus so reducing sperm penetrability and transport.
- The levonorgestrel intrauterine system (LNG-IUS) has little effect on ovarian activity but causes marked endometrial atrophy, which prevents implantation if ovulation and fertilization occur.

Progestogen-only pill

- Unlike the COCP, the POP needs to be taken continuously.
- Medium-dose pills (e.g. containing desogestrel) inhibit ovulation in 99% of cycles, but lower-dose pills inhibit ovulation in less than one half of cycles, relying on the cervical mucus effect for contraception.

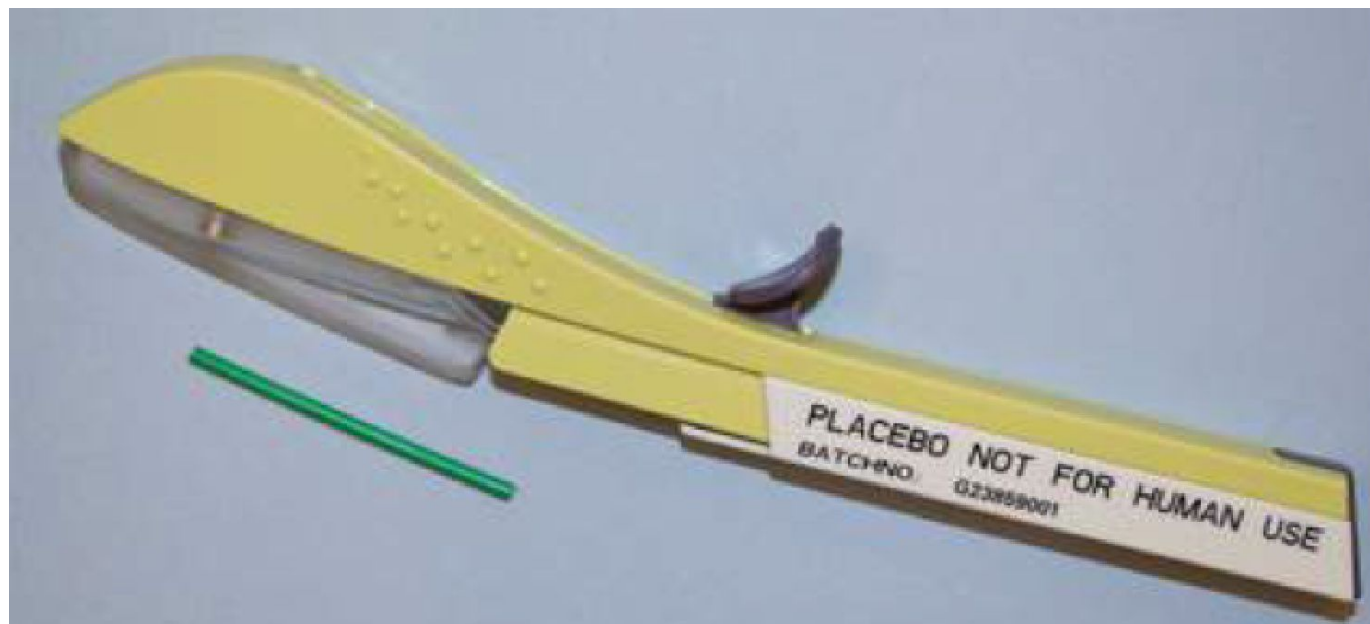


- Side-effects of all POPs include possible irregular bleeding, persistent ovarian follicles (simple cysts) and acne.
- If a POP is missed then the woman should continue taking the POP and use extra precautions (e.g. condoms) for the next 48 hours until the progestogen effect on the mucus is built up.
- If unprotected sex occurs during this time, then emergency contraception is required.

Implant

- A single rod (Nexplanon®) containing the progestogen etonorgestrel is the currently available method in the UK.
- Nexplanon® contains 68 mg of 3-keto-desogestrel (a metabolite of desogestrel) providing contraception for 3 years.

Nexplanon.



- Nexplanon® is a flexible rod, similar in size to a match stick (40 mm × 2 mm) and is inserted subdermally 8 cm above the medial epicondyle, usually of the non-dominant arm.
- Insertion is conducted under local anaesthesia using a specially designed insertion device.
- poor insertion technique can still result in deep insertion with consequent difficult removal, so insertion should only be conducted by clinicians who have undertaken appropriate training.



- The implant is not usually visible, but should be easily palpable. Nexplanon® contains a small quantity of barium, which permits it to be visualized by X-ray. It can also be localized using low-frequency ultrasound probes, which can help aid removal of implants that are not easily palpable.
- Once inserted, there is no need for any routine follow-up until the device is due for replacement or the user wishes it removed.



- Nexplanon® releases a steady low dose of progestogen (similar to levels of a (POP)).
- **After removal, serum levels of etonorgestrel levels are undetectable within 1 week, and fertility is restored immediately after removal.**



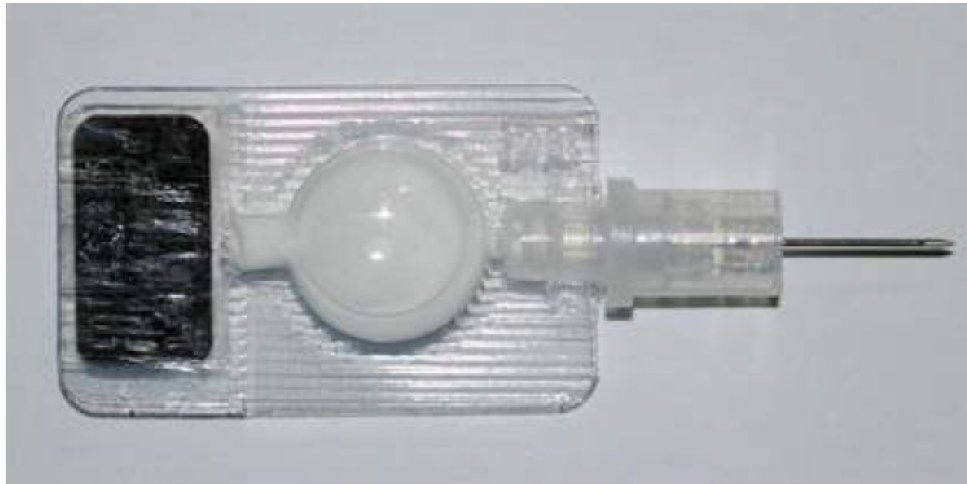
Progestogen-only injectable:

- The most commonly used injectable worldwide is a depot injection of medroxyprogesterone acetate, which can be administered intramuscularly as the formulation Depoprovera® (150 mg) or subcutaneously as the micronized lower-dose formulation of Sayana press® (104 mg).
- Both intramuscular and subcutaneous preparations have similar features: same mode of action (inhibition of ovulation), same efficacy, similar injection interval (every 12–14 weeks) and similar bleeding pattern (over 50% amenorrhoea rates at 1 year).
- The injectable is the only hormonal method that may delay return of fertility after discontinuation.

In some cases it may take up to 1 year after the last injection for ovulation to return.

- There is no permanent impairment of fertility but this delay makes the injectable an inappropriate method for women wishing short-term contraception.

Sayana®-press single-dose container



Depoprovera®





- Both the intramuscular and subcutaneous preparation may cause weight gain in a minority of women and loss of bone mineral density (BMD) (5% loss of BMD at lumbar spine) in the first few years of use.
- This bone marrow loss has not been associated with osteoporotic fractures and appears to be reversible on stopping.
- So Condom use in addition to the injectable should also be encouraged to protect against transmission or acquisition of HIV.

Progestogen-releasing intrauterine system:

- The currently available intrauterine systems release the progestogen levonorgestrel into the uterus.
- There are two LNG-IUS that are available in Europe and the USA. The 52 mg LNG-IUS (Mirena®) is licensed for 5 years for contraceptive use (but if inserted in women 45 years or older, may be used for contraception until the menopause) and the 13.5 mg LNG-IUS (known as Jaydess® in Europe and Skyla® in the USA) is licensed for 3 years for contraceptive use.
- The 13.5 mg LNG-IUS has a slightly narrower insertion device and it has a shorter frame, which may make insertion easier in young or nulliparous women.



- The LNG-IUS works by exerting a potent hormonal effect on the endometrium, which prevents endometrial proliferation and implantation.
- Its progestogenic effect on thickening the cervical mucus also impedes entry of sperm.
- The LNG-IUS does not prevent ovulation. In the first few months of use, many women experience unpredictable bleeding.
- Women should be advised that this usually improves with time and many women will eventually have lighter or absent periods.
- Provision of quality information about side-effects in advance of fitting a LNG-IUS is important to reduce unnecessary discontinuation rates



- Reported side effects of the LNG-IUS include acne, breast tenderness, mood disturbance and headaches.
- The most notable non-contraceptive benefit of the 52 mg LNG-IUS is that of reducing HMB (reduced by 90% at 12 months).
- It is more effective than oral treatments, such as norethisterone, the COCP and tranexamic acid, at reducing menstrual blood.
- It is also effective for treating dysmenorrhoea, pain associated with endometriosis and adenomyosis and protecting the endometrium against hyperplasia.



Intrauterine contraception

- Include the copper intrauterine device Cu-IUD and the LNG-IUS.
- The Cu-IUD duration of use is between 3 and 10 years, depending on the device used and age of woman at insertion.
- If a woman has a Cu-IUD inserted at 40 years or above, it can be left *in situ* until the menopause. LNG-IUS inserted at 45 years or over, the device can be left for contraceptive purposes until the menopause.
- There are a number of Cu-IUDs available and they vary in size, shape, copper content and duration of use.
- Most consist of a plastic frame with copper wire wound around the stem and some may have copper on the arms of the device. The LNG-IUS consists of an elastomere frame with a reservoir on the stem containing levonorgestrel.

- With both Cu-IUD and LNG-IUS, threads protrude through the cervical canal into the upper vagina to permit easy removal.
- Evidence suggests that the Cu-IUD and the LNG-IUS do not cause a delay in return to fertility or increase the risk of infertility.

- **Mode of action**

- IUDs stimulate an inflammatory reaction in the uterus. The concentration of macrophages and leucocytes, prostaglandins and various enzymes in both uterine and tubal fluid increase significantly. It is thought that these effects are toxic to both sperm and egg and interfere with sperm transport. If a healthy fertilized egg reaches the uterine cavity, implantation is inhibited.

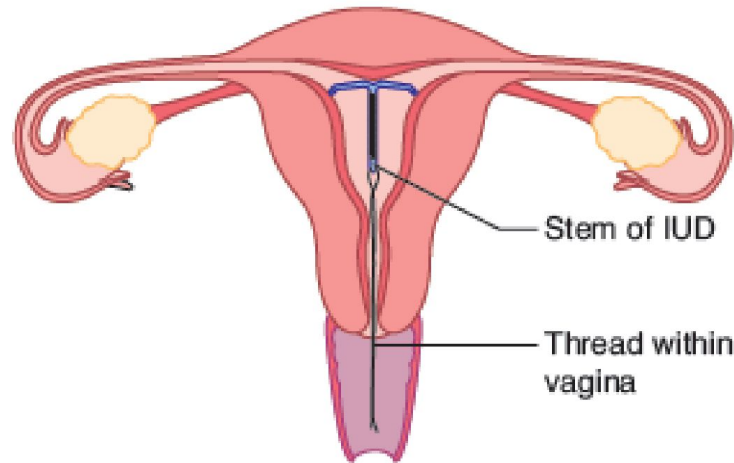


Figure 7.10 Intrauterine device (IUD) in the uterine cavity.



Figure 7.12 Hormone-releasing intrauterine system – Mirena. Photograph courtesy of Bayer Schering Ltd.

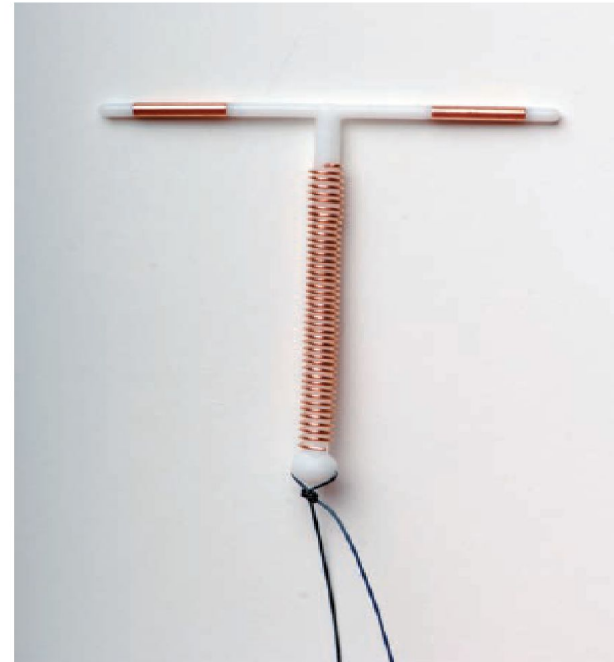


Figure 7.11 Copper bearing IUD with copper on stem and arms. Photograph courtesy of Durbin Ltd.

	Copper IUD	Mirena
Failure rate in first year of use	0.8%	0.1%
Mode of action	Toxic effect on both sperm and egg, i.e. acting prior to fertilization	Local hormonal effect on the cervical mucus and endometrium
Duration of use	10 years	5 years
Effect on menstrual cycle	Periods can become heavier with more pain	Periods become irregular but much lighter. Women often become amenorrhoeic
Menstrual spotting	Often more days of spotting before and after periods	Erratic spotting very common initially but usually settles
Hormonal side effects	None	May cause greasier skin and acne, breast tenderness and mood swings. Symptoms often settle with time
Therapeutic benefits	None	Helps heavy and painful periods. Can use as part of a hormone replacement therapy regimen
Average UK cost	£10	£80

Bleeding pattern with IUD

- Although women with the LNG-IUS tend to experience lighter, less painful menses, women using the Cu- IUD may experience more painful or heavier menses.
- Women using an IUD should be informed that their overall risk of ectopic pregnancy is much reduced compared with women who are using no contraception.
- However, if a pregnancy does occur with an IUD *in situ* then the 'relative' risk of that pregnancy being ectopic is higher.

- If women become pregnant with an IUD *in situ*, an ultrasound scan should be conducted to exclude ectopic pregnancy. It is generally advisable that IUDs should be removed before 12 weeks' gestation in view of the greater risk of miscarriage, preterm delivery, septic abortion and chorioamnionitis if the device is left *in situ*.
- Although there is a theoretical concern about teratogenicity if a pregnancy is exposed to the LNG-IUS, to date no birth defects have been reported in the small number of cases exposed.

Insertion of IUD

An IUD can be fitted at any point in the cycle provided there is no risk of pregnancy. Insertion is associated with the following risks:

Perforation.

- is rare 1 in 1,000. Factors associated with an increased risk of perforation include relative inexperience of clinician, breastfeeding and being less than 6 months postpartum.
- Usually the woman presents with 'missing threads' and a history of severe pain following insertion. An ultrasound of the uterus confirms no IUD is seen in the uterus and an abdominal X-ray shows an IUD.
- Laparoscopic retrieval of the device from the pelvis (often attached to omentum) is possible in most cases, but laparotomy may rarely be required to remove the IUD.

Expulsion: 1 in 20 IUD devices will be expelled in the first 3 months after insertion. After this, the risk of expulsion diminishes.

Infection:

 The overall risk of pelvic infection in the first 3 weeks following insertion of an IUD is low 1%. Thereafter, the risk of infection is the same as women not using contraception.

- In most cases, the IUD can be left *in situ* and antibiotic therapy commenced.
- In addition, women with symptoms of pelvic infection with an IUD should be tested for sexually-transmitted infections.



Missing' threads :may indicate pregnancy, expulsion or perforation. However, it is often the case that the threads are merely sitting in the cervical canal or uterus.

- A pregnancy test should be performed and emergency contraception/alternative contraception provided until the IUD can be confirmed to be *in situ*, either by visualization of the threads on speculum examination or an ultrasound confirming presence of the IUD within the uterus.

Contraindications for intrauterine contraceptive device:

- Current STD or PID .
- Malignant trophoblastic disease.
- Unexplained vaginal bleeding (before assessment).
- Endometrial and cervical cancer.
- Known malformation of the uterus or distortion of the cavity (fibroids).
- Copper allergy (but could use a Mirena).



Route of administration of contraceptive and duration

Route of currently available contraceptives	Duration
Oral CHC and progestogen	24 hours
Transdermal CHC	7 days
Vaginal ring CHC	21 days
Progestogen-only injectable	14 weeks
Progestogen-only implant	3 years
Cu- IUD	3 years, 5 years, 10 years or more
LNG-IUS	3 years or 5 years or more

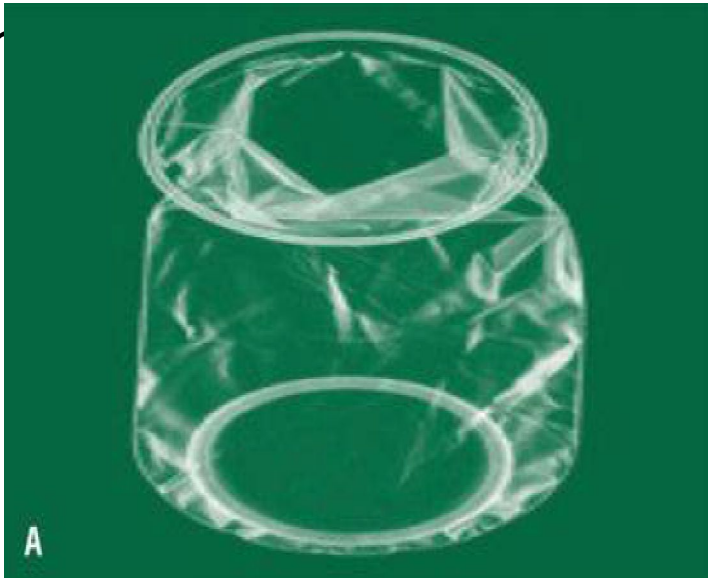
Barrier contraception

Condoms

Male condoms are cheap and widely available. They protect against STIs including HIV. They are the only reversible male method.

Typical failure rates are in the region of 24% since they rely on the user to put it on it correctly, before penetration and before every act of sex.

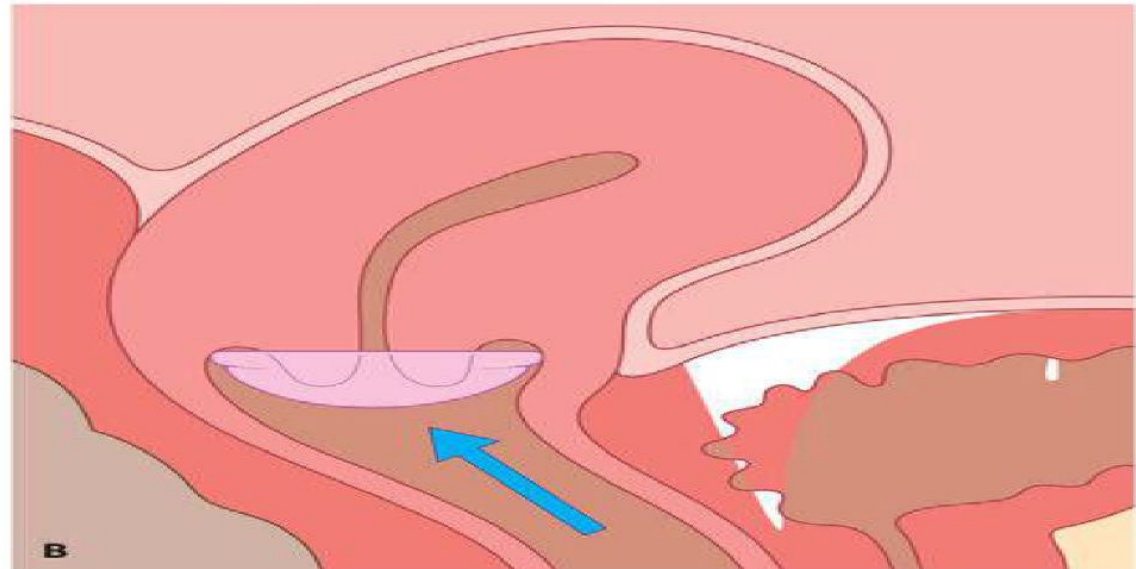
The female condom is a lubricated polyurethane condom that is inserted into the vagina. It is also





Diaphragm and cap

- These are latex or non-latex devices that are inserted into the vagina to prevent passage of sperm to the cervix .
- Caps and diaphragms are often used in conjunction with a spermicide.
- Disadvantages are that women need to be taught how to insert and remove the device and typical failure rates in the region of 18% are reported.
- In some women their use may be associated with increased vaginal discharge and urinary tract infections.



A: The cap; **B:** correct sitting of a cap

Spermicides: alone is not recommended for prevention of pregnancy as it is of low effectiveness. Nonoxynol 9 (N-9) is a spermicidal product sold as a gel, cream, foam, sponge or pessary for use with diaphragms or caps. Some data have suggested that frequent use of N-9 might increase the risk of HIV transmission. It is therefore no longer recommended for women who are at high risk of HIV infection.

Female sterilization

- This is a permanent method of contraception that prevents sperm reaching the oocyte in the Fallopian tube.
- It can be performed by (1) laparoscopy, (2) hysteroscopy or (3) laparotomy (e.g. at caesarean section)

Laparoscopic sterilization

- Laparoscopic sterilization most commonly occludes the Fallopian tube with filshie clips.
- Effective contraception should be used until the next menses after the procedure, due to the risk of pregnancy from implantation of an early fertilized egg in the same cycle as sterilization.
- Women who pose a higher surgical risk due to previous abdominal surgery, for example, may be suited to the hysteroscopic approach.
- Since sterilization results in permanent loss of fertility and involves a surgical procedure, it is important that valid consent is obtained.

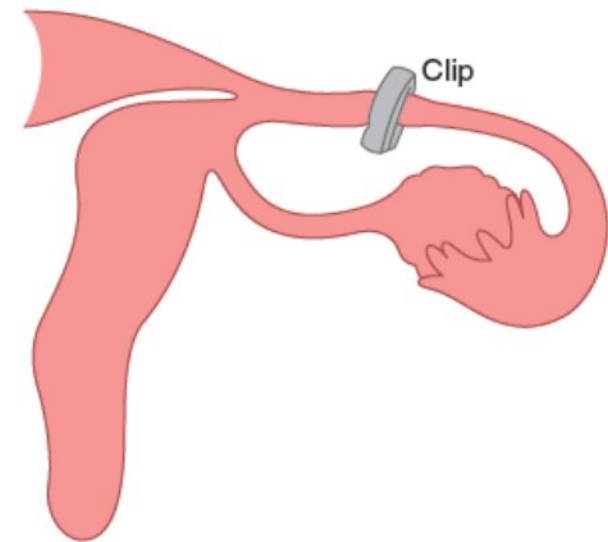
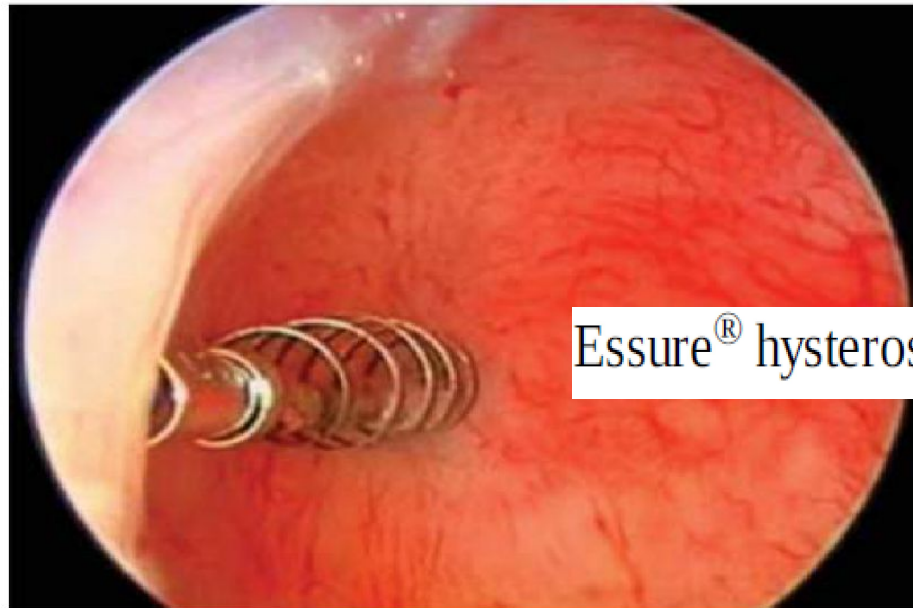
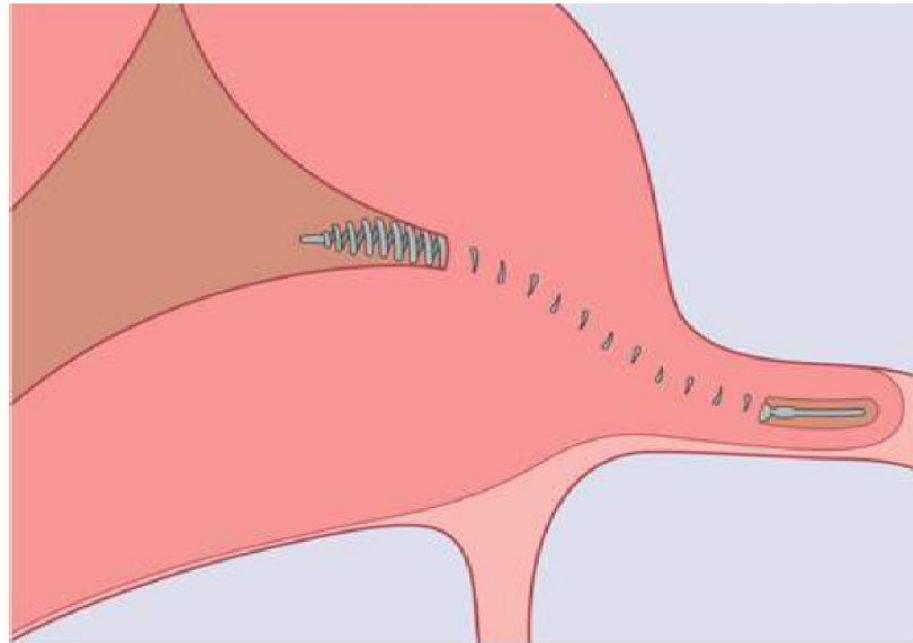


Figure 7.15 Female sterilization.

Hysteroscopic sterilization

- This has the advantage that it can be performed as an outpatient procedure without general anaesthesia.
- Microinserts (Essure®), which are expanding springs (of 2 mm diameter and 4 cm in length) made of titanium, steel and nickel-containing Dacron fibres, are inserted into the tubal ostia via a hysteroscope.
- These induce fibrosis within the cornual section of each Fallopian tube over the following 3 months.
- Contraception is required during the 3 months and can only be discontinued once correct placement of the inserts are confirmed by X-ray imaging or ultrasound.



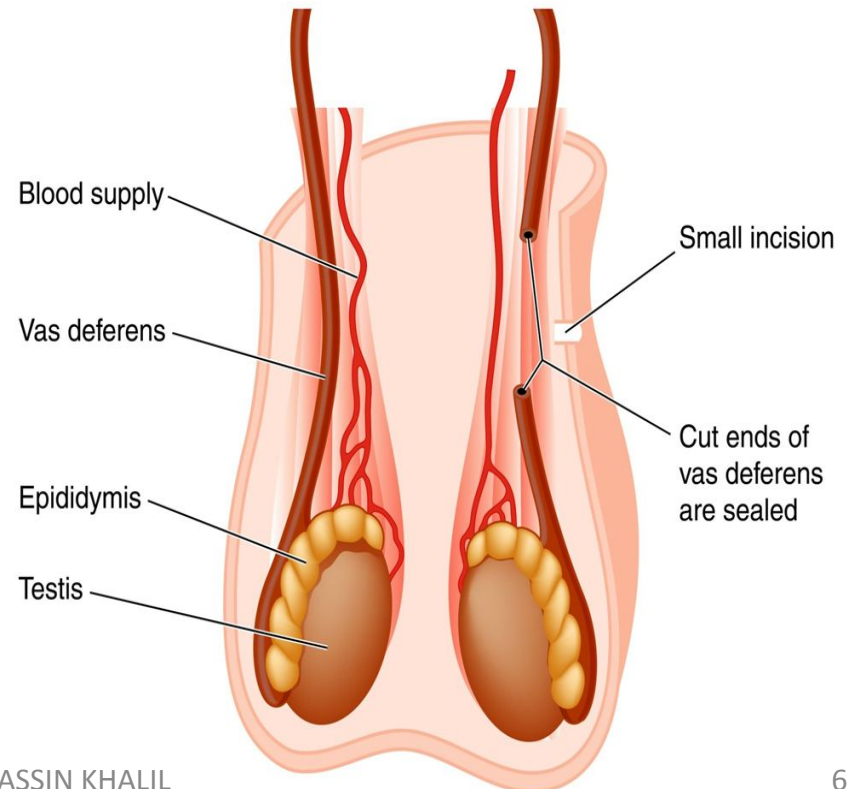
Essure[®] hysteroscopic sterilization.

Vasectomy

- This is the technique of interrupting the vas deferens to provide permanent occlusion. Involves the division of the vas deference on each side to prevent the release of sperm during ejaculation.
- The so-called 'no scalpel' vasectomy involves a puncture wound in the skin of the scrotum under local anaesthesia to access and then divide and occlude the vas using cautery.
- There is a small risk of a scrotal haematoma and infection with the procedure.
- Postvasectomy semen analysis should be conducted at 12 weeks to confirm the absence of spermatozoa in the ejaculate. Alternative contraception should be used until azoospermia is confirmed.
- The failure rate is significantly less than female sterilization at approximately 1 in 2,000.

Complications :

- Immediate : bleeding, haematoma, infection.
- Small lump(sperm granuloma) at the site of incision.
- Anti-sperm antibodies prevent pregnancy after reversal.
- Scrotal pain
- Possible association with prostatic
- Types : ligation or clips, excision, no-scalpel vasectomy and silicone plugs/ sclerosing agents.



Advice to women considering sterilization

- Method is considered as irreversible.
- Failure rate 1:200 for laparoscopic, 1:500 for hysteroscopic (comparable to long-acting reversible methods).
- Risks and complications (laparoscopic 1:1,000 risk of trauma to bowel, bladder or blood vessels).
- Vasectomy is safer, quicker, safer and with less morbidity.
- High proportion of women regret sterilization. Risk factors are age under 30 years, nulliparity, recent pregnancy (birth, abortion, miscarriage) and relational issues.
- Does not protect against STIs.
- Effective contraception is required until the menstrual period following laparoscopic procedure or 3 months following hysteroscopic procedure.
- Pregnancy following female sterilization is rare but if it does occur there is an increased risk of ectopic.
- Reversal of sterilization is a highly-skilled procedure to obtain tubal reanastomosis. It cannot be performed after hysteroscopic sterilization, and if it is successfully conducted after laparoscopic sterilization, then it is associated with an increased risk of ectopic pregnancy.



Fertility awareness-based methods

- Formerly known as ‘natural family planning’, FAB rely on the signs and symptoms that reflect the physiological changes that occur during the menstrual cycle that define the fertile period, with avoidance of intercourse at that time.
- Use of FAB requires motivation and a regular menstrual cycle, and so cannot be used for women at extremes of reproductive age.
- Typical failure rates are high.
- The method depends on the use of one or more of the following indicators to enable avoidance of intercourse during the fertile days.

- **Calendar or rhythm method** Fertile days are calculated based upon the cycle length recorded over at least six cycles. First fertile day = shortest cycle minus 20. Last fertile day = longest cycle minus 10. For women with a 28-day cycle this equates to abstinence for 10 days in each cycle (i.e. day 8–18).
- **Temperature method** This relies on the increase in basal body temperature (0.2–0.4C) produced by the rise in progesterone following ovulation. Daily temperatures must be measured using the same route. Infection, exercise and some medications can affect body temperature and interfere with this method.
- **Cervical palpation:** At midcycle, the cervix rises 1–2 cm and feels softer and moist.



Cervical mucus method ; Mucus on toilet

tissue after wiping the vulva can be examined for consistency. Midcycle 'fertile' mucus due to rising oestradiol levels is clear, watery and slippery rather like raw egg white. Following ovulation, progesterone renders it thick and opaque. Semen in the vagina may make it difficult to recognize the mucus.

- **Personal fertility monitor**: This hand-held monitor analyses disposable urine dipsticks that record the presence of metabolites of oestrogen and LH in the urine.
- It recognizes urinary oestrogen concentrations corresponding to the midfollicular phase of the cycle and the preovulatory LH peak, so that the beginning and end of the fertile phase can be identified.
- A red light indicates fertile phase (risk of conception) and green, infertile. Users need to perform urine dipstick tests on early morning urine. A red light is usually shown for 6–10 days in the cycle.

Withdrawal :

- Is widely used and require no medical supervision.
- The penis is withdrawn from the vagina just before ejaculation.
- Had higher failure rate because the pre-ejaculatory fluid may contain sperm .

Lactational amenorrhoea

- If a mother is within the first 6 months postpartum, is amenorrhoeic and is fully or nearly fully breastfeeding, then the risk of pregnancy is about 2%.
- After 6 months, or if menses occur or breastfeeding reduced, then another method of contraception must be used.

Emergency contraception

- All women deserve a second chance to prevent an unintended pregnancy.
- The Cu-IUD is the most effective method of emergency contraception (EC) available (failure rate 1 in 1,000) and should ideally be offered as first choice to women.
- When used for EC, its effect on the endometrium is thought to prevent implantation if fertilization has occurred.
- The Cu-IUD can be removed once pregnancy has been excluded or can be left in place for ongoing contraception.
- Since the blastocyst will implant between 6 and 10 days after fertilization, an emergency Cu-IUD can be inserted up to 5 days after the unprotected sex or 5 days after predicted ovulation.

- There is no evidence that the LNG-IUS is effective as EC and should therefore not be used for this purpose.
- The two oral methods of EC that are licensed for use are levonorgestrel (LNG 1.5 mg) and the progesterone receptor modulator ulipristal acetate (30 mg).
- LNG appears effective up to 96 hours after unprotected sex and ulipristal acetate for up to 120 hours.
- Both methods work by delaying ovulation, so that any sperm present in the reproductive tract will have lost the ability to fertilize the oocyte once it is eventually released.
- Oral EC is much less effective than the Cu-IUD for EC and is estimated to prevent only two-thirds of pregnancies

Emergency contraception

- The most effective method of EC is an IUD (about 99% effective).
- An IUD can be inserted up to 5 days after ovulation for EC.
- Ulipristal acetate (UPA) or levonorgestrel (LNG) are available as oral methods of EC.
- UPA can be given within 120 hours of unprotected intercourse.
- LNG can be used within 96 hours of unprotected intercourse.
- Effective ongoing contraception should be started after EC.



Opportunities to provide contraception

- There are important opportunities to discuss and provide effective contraception when women present to gynaecology clinics.
- These include presentation to a general gynaecology clinic, presentations for EC, presentation to request an induced abortion and also antenatally or after childbirth.
- The antenatal/postpartum period is a particularly important opportunity since 95% of postpartum women want to avoid pregnancy in the next 12 months, and in women who are not fully breastfeeding, fertility may return at 1 month postpartum.

Conclusion:

- Most types of contraception are safe.
- Most types of contraception are tolerable.
- LARC are the best contraception which is not dependant on patients use.



COLLEGE
OF
MEDICINE

THANK YOU